

Custom iPod Project

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1 Design Brief

1.1 Overview

The primary objective of this device is to design and implement a portable, battery-powered digital audio player capable of high-fidelity playback of lossless audio formats. This design will focus on only local playback of audio files, rather than streaming from cloud services.

This device will be implemented using an STM32 microcontroller, and will focus on repairability, transparency, and reproducibility.

1.2 Motivation

Most portable audio players are either a smartphone with compromised audio stages (due to other requirements of those devices), or premium digital audio players that are expensive and proprietary. This project aims to demonstrate and construct a purpose-built embedded platform that can achieve comparable objective performance using modern components while remaining open and reproducible.

1.3 Objectives

1.3.1 Primary Objectives

Below are the primary objectives of the device.

1. It should be portable and battery-powered
2. It should be capable of playing back lossless audio formats
3. It should have a high-dynamic range
4. It should have low output noise
5. It should have low harmonic distortion
6. It should have support for low- to mid-power headphones and audio devices
7. It should have long battery life

1.3.2 Secondary Objectives

1. It should have the ability to function as a USB DAC
2. It should have expandable storage
3. It should have easy to update firmware
4. It should be completely open-source
5. It should be completely user-repairable

1.4 Requirements

1.4.1 Functional Requirements

1. The device shall be capable of playing all popular audio formats
 - (a) WAV
 - (b) FLAC
 - (c) MP3
2. It shall support pulse-control modulation (PCM) up to at least 24-bit/192 kHz
3. It shall decode files from removable storage
4. It shall maintain playback location between sessions or after power loss.
5. It shall display metadata about the currently playing file
 - (a) Title/Artist/Album
 - (b) Duration/position in file
 - (c) Album cover/file icon
6. It shall provide hardware volume control
7. It shall provide information on the state of the device
 - (a) Battery monitoring
 - (b) Storage capacity and usage
8. It shall utilise the current standard connectors/ports where applicable
 - (a) The charging and data transfer shall use a USB Type-C connection
 - (b) The audio output shall use a standard 3.5mm audio jack.

1.4.2 Performance Requirements

The device shall have at minimum the following specifications

Table 1: Performance requirements for the device	
Parameter	Target
Frequency response	20 Hz-20 kHz ± 0.2 dB
THD+N (total harmonic distortion)	$< 0.001\%$
Dynamic range	> 120 dB
SNR (signal-to-noise ratio)	> 120 dB
Crosstalk	< -100 dB
Output impedance	$< 1\Omega$
Headphone output	> 500 mW @ 32Ω balanced
Idle battery life	> 20 h
Playback battery life	> 10 h

1.4.3 Interface Requirements

1. The interface shall match or exceed the follow specifications of the iPod Classic 7th Gen display
 - (a) 2.5in (63.5 mm) (diagonal) colour LCD with LED backlight
 - (b) 320x240 pixel resolution at 163 PPI

1.5 Design Considerations

1.5.1 Audio Integrity

1. The device shall separate analogue and digital ground planes with a controlled connection
2. The device shall isolate high-speed digital traces from sensitive analogue circuitry
3. The device shall use differential routing for DAC outputs where appropriate
4. The device shall minimise loop areas and return-path discontinuities

1.5.2 Power

1. The device shall have separate analogue and digital regulators
2. The device shall have low-noise references
3. The device shall have high PSRR LDOs
4. The device shall have sufficient decoupling

1.5.3 EMC

1. The device shall shield crystal oscillators
2. The device shall have controlled impedance USB routing
3. The device shall have ground stitching around analogue sections

1.6 Verification

In order to verify device performance, the following tests must be conducted

1. Frequency response
2. THD+N
3. SNR
4. Dynamic range
5. Crosstalk
6. Output impedance
7. Maximum output power
8. Battery runtime
9. Power consumption
10. Thermal performance

2 Research

2.1 DAC Chip

1. PCM1794ADBR - High-quality 24-bit DAC with 127-132 dB dynamic range.

2.2 Display

1. 1.69 LCD Display
2. NHD-2.4-240320DB-CTXP

2.3 MCU

1. STM32G474CBT6:
 - 1x SAI
 - 4x SPI
 - 4x I2C
 - 512 Kb Flash
 - 92 Kb SRAM
2. STM32U535CBU6

2.4 Batteries

1. 3V7 2000mAh